



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problemsolving skills through emerging technologies.

Session 2025-2026

Vision: <i>To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration</i>	Mission: <i>To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.</i>
---	---

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

(Signature and Date in Handwritten)



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problemsolving skills through emerging technologies.

Session	2025-26 (EVEN)	Course Name	JAVA FSD Lab
Semester	4th	Course Code	23ADS1407
Roll No	120	Name of Student	Shrusti katakwar

Practical Number	09
Course Outcome	• Understand how Java applications connect to databases using JDBC. • Learn to execute SQL queries from a Java program. • Develop skills to perform CRUD operations (Create, Read, Update, Delete). • Gain knowledge of database drivers and connection handling.
Aim	WAP to demonstrate database connectivity using JDBC
Problem Definition	WAP to demonstrate database connectivity using JDBC



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

Theory (100 words)	JDBC (Java Database Connectivity) is a standard API provided by Java that enables Java applications to interact with databases. It allows a program to connect to a database, send SQL queries, and process the results. JDBC acts as a bridge between the Java application and the database. It provides methods and interfaces to establish a connection, execute queries, and handle data returned from the database. Advantages of JDBC • Platform-independent (Write Once, Run Anywhere) • Supports multiple databases • Provides simple and easy database connectivity • Enables execution of SQL queries from Java programs • Supports transaction management Applications of JDBC • Banking systems • Student management systems • Online registration systems • E-commerce applications
-----------------------	---

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problemsolving skills through emerging technologies.

Procedure and Execution (100 Words)	Algorithm : 1. Start 2. Import required packages 3. Load JDBC driver 4. Establish connection with database 5. Create statement object 6. Execute SQL query 7. Store result in ResultSet 8. Display the data 9. Close connection 10. ☐ Stop
--	---



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

Code:

```
Main.java JDBCDemo.java x
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class JDBCDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String user = "root";
        String password = "root"; // Change if different
        try {
            // Step 1: Establish Connection
            Connection con = DriverManager.getConnection(url, user, password);
            System.out.println("Connected to Database Successfully!");

            // Step 2: Create Statement
            Statement stmt = con.createStatement();

            // Step 3: Execute Query
            ResultSet rs = stmt.executeQuery("sql: SELECT * FROM student");

            // Step 4: Process Result
            while (rs.next()) {
                System.out.println(
                    "ID: " + rs.getInt(columnLabel: "id") +
                    ", Name: " + rs.getString(columnLabel: "name")
                );
            }
        }
    }
}
```



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

```
);  
  
    }  
    // Step 5: Close Connection  
    con.close();  
} catch (Exception e) {  
    e.printStackTrace();  
}  
}  
}
```

Output:

```
Run JDBCDemo x  
"C:\Program Files\Java\jdk-25\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1  
Connected to Database Successfully!  
ID: 1, Name: Rahul  
ID: 2, Name: Amit  
Process finished with exit code 0
```

Output Analysis

- The program connects successfully to the database.
- It retrieves all records from the student table.
- Each record is displayed with roll number, name, and marks.



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

	If connection fails, an exception message is shown.
--	---

Link of student Github profile where lab assignment has been uploaded	https://github.com/Shrusti-Katakwar/shrusti_java/commit/2f8b6b7e9d2ddd91712120d4fc66d3e0d2c93268
Conclusion	- JDBC enables seamless interaction between Java applications and databases. - It simplifies executing SQL queries and retrieving results. - Proper exception handling ensures reliability. - This program demonstrates basic database connectivity and data retrieval.



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



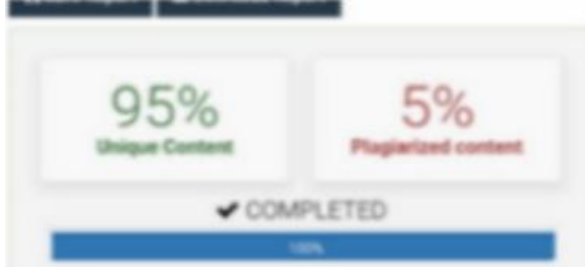
Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

Plag
Report
(Similarit
y index <
12%)





Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problemsolving skills through emerging technologies.

Date	28/03/26
------	----------

Paste here Faculty signed page

